Problem 1 (4 points) Let $f(x,y) = \ln(y^2 - x - 2)$.

a) (3 points) Find an equation for the tangent plane to f at $(1,2)$.

$f(1,2) = \ln(1) = 0$.

$f_x = \dfrac{-1}{y^2 - x - 2}$, so $f_x(1,2) = -1$.

$f_y = \dfrac{2y}{y^2 - x - 2}$, so $f_y(1,2) = 4$.

The tangent plane is $z = -(x-1) + 4(y-2)$.

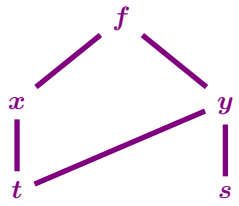
b) (1 point) Use your answer to part a) to approximate $f(0,3)$.

$f(0,3) \approx -(0-1) + 4(3-2) = 5$.

Problem 2 (4 points) A function $f(x, y)$ has inputs $x = x(t)$ and $y = y(t, s)$. You are given the following information:

$$\frac{\partial f}{\partial x} = 1 \qquad \frac{\partial y}{\partial t} = 2 \qquad \frac{\partial y}{\partial s} = 5 \qquad \frac{\partial f}{\partial s} = -15 \qquad \frac{\partial f}{\partial t} = -10.$$

Find $\dfrac{\partial x}{\partial t}$.



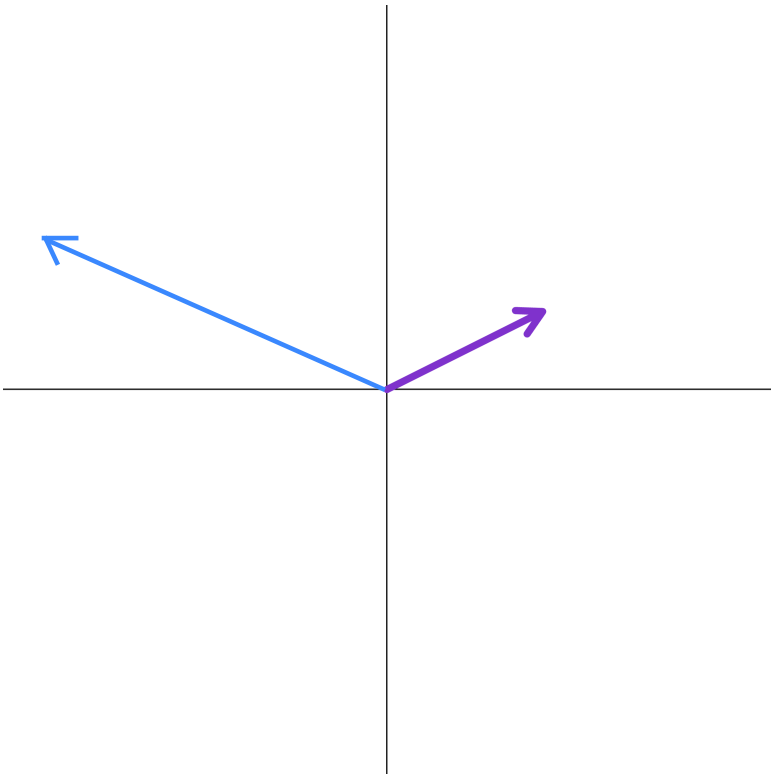
By the multivariable chain rule, $\dfrac{\partial f}{\partial s} = \dfrac{\partial f}{\partial y}\dfrac{\partial y}{\partial s}$, so $-15 = 5\dfrac{\partial f}{\partial y}$ and $\dfrac{\partial f}{\partial y} = -3$. Then

$$\frac{\partial f}{\partial t} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} \implies -10 = \frac{\partial x}{\partial t} - 6,$$

so $\dfrac{\partial x}{\partial t} = -4$.

Problem 3 (3 points) Drawn below is a unit vector $\vec{u}$. Draw a vector $\vec{v}$ so that $\|\vec{v}\| = 2$ and $\vec{u} \bullet \vec{v} = -1$. Indicate any relevant angles and lengths in your drawing.

$\vec{u} \bullet \vec{v} = \|\vec{u}\|\,\|\vec{v}\|\cos(\theta) = 2\cos(\theta) = -1$, so $\theta = \dfrac{2\pi}{3}$. $\vec{v}$ has twice the length of $\vec{u}$ at an angle of 120° from $\vec{u}$ on either side.



Problem 4 (10 points) Let P be the unique plane that passes through the points $(1,2,1)$, $(2,3,4)$, and $(3,1,1)$.

a) (4 points) Find an equation for P.

Two vectors in P are $\langle 1,1,3\rangle$ and $\langle 2,-1,0\rangle$, so a normal vector is $\langle 1,1,3\rangle \times \langle 2,-1,0\rangle = \langle 3,6,-3\rangle$, or $\langle 1,2,-1\rangle$. Then P is $(x-1)+2(y-2)-(z-1)=0$ or $x+2y-z=4$.

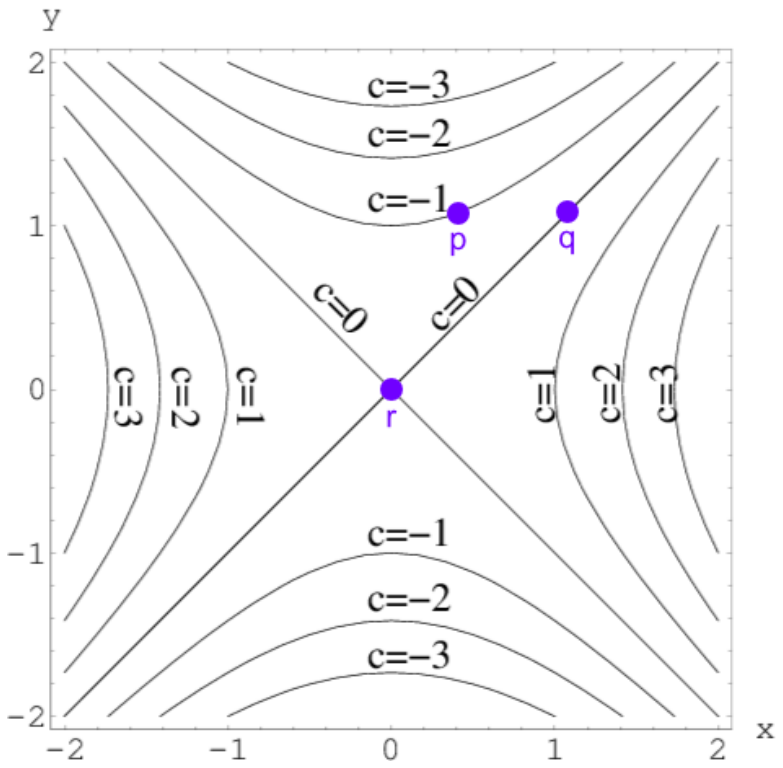
b) (2 points) Let L be the line passing through $(3,3,-1)$ and $(2,5,-4)$. Find a formula for L.

$L(t) = \langle 3,3,-1\rangle + t\langle -1,2,-3\rangle$.

c) (4 points) Find the intersection between P and L, or show they do not intersect.

Plugging $x=3-t$, $y=3+2t$, $z=-1-3t$ into the formula for P results in $10+6t=4$, so $t=-1$. The intersection point is then $(4,1,2)$.

Problem 5 (9 points) The following graph shows the level curves of a function $f(x, y)$, with marked points $\vec{p}$, $\vec{q}$, and $\vec{r}$.



a) (1 point) What is $f(\vec{p})$? (No justification required.)

It's on the $c = -1$ level curve, so -1.

b) (2 points) Is $f_y(\vec{q})$ positive, negative, or zero? Explain with a sentence.

Negative, since moving up from $\vec{q}$ reaches lower-valued level curves.

c) (3 points) Is $f_x(\vec{q})$ larger than, smaller than, or equal to $f_x(\vec{p})$? Explain.

Larger, since the $c = 2$ level curve is closer to $\vec{q}$ then the $c = 1$ level curve is to $\vec{p}$.

d) (3 points) Explain why f has a critical point at $\vec{r}$. Is it a local maximum, local minimum, or a saddle point?

The level curves cross at $\vec{r}$, so the gradient there can't be perpendicular to both of them unless it's $\vec{0}$, which means $\vec{r}$ is a critical point. f increases in some directions and decreases in others away from $\vec{r}$, so it's a saddle point.

Problem 6 (6 points) Let $g(x,y) = x^2 + xy^2 + \sin(\pi y)$.

a) (2 points) Find $\nabla g(-2,2)$.

$g_x = 2x + y^2$

$g_y = 2xy + \pi\cos(\pi y)$

$\nabla g(-2,2) = \langle 0, \pi - 8\rangle$.

b) (4 points) Let L be the line that is tangent to the graph of g at $(-2,2,g(-2,2))$ and that lies in the plane $x = -2$. Find a formula for L.

$g(-2,2) = -4$. In the plane $x = -2$, L moves in the y-direction with slope $g_y(-2,2) = \pi - 8$, so

$$L(t) = \langle -2, 2, -4\rangle + t\langle 0, 1, \pi - 8\rangle .$$

Problem 7 (7 points) A small technology company selling x laptops, y tablets, and z smartphones per day finds that they make a daily profit of

$$P(x, y, z) = 3x + 7y + 5z + yz - xy$$

in hundreds of dollars. Currently, they sell 5 laptops, 1 tablet, and 1 smartphone per day.

a) (3 points) Find $P_x(5, 1, 1)$ and write a sentence interpreting its value.

$P_x = 3 - y$, so $P_x(5, 1, 1) = 2$. Selling one additional laptop per day increases daily profit by about \$200.

b) (4 points) Each device costs the company \$100 to produce, and they have a budget of \$1100 to spend on additional daily production. If they increase production in the direction that increases profit the fastest and spend their entire budget, how many additional laptops, tablets, and smartphones should they produce?

The direction of fastest increase is

$$\nabla P(5, 1, 1) = \langle 3 - y, 7 + z - x, 5 + y \rangle = \langle 2, 3, 6 \rangle .$$

They can produce 11 additional devices, and $2 + 3 + 6 = 11$, so they should produce 2 additional laptops, 3 additional tablets, and 6 additional smartphones.

Problem 8 (10 points) Let $f(x,y) = x^3 + 3y^2 - 6xy$.

a) (4 points) Find and classify the critical points of f.

$f_x = 3x^2 - 6y = 0$ and $f_y = 6y - 6x = 0$ results in $y = x$ and $x^2 = 2x$, so the critical points are $(0,0)$ and $(2,2)$. With $f_{xx} = 6x$, $f_{yy} = 6$, and $f_{xy} = -6$, we have $D = 36x - 36$.

- At $(0,0)$, $D = -36 < 0$: saddle point.
- At $(1,1)$, $D = 36 > 0$ and $f_{yy} = 6 > 0$: local minimum.

b) (6 points) Find the minimum and maximum values of f along the line $y = -x$ between $x = -10$ and $x = 1$.

We optimize $f(x,-x) = x^3 + 9x^2$. If $g(x) = f(x,-x)$, then $g'(x) = 3x^2 + 18x = 3x(x+6)$, which is zero at $x = 0$ and $x = -6$. Checking critical points and endpoints:

$$g(-10) = -100, \quad g(-6) = 36(-6+9) = 108, \quad g(0) = 0, \quad g(1) = 10.$$

The maximum is 108 at $(-6,6)$ and the minimum is -100 at $(-10,10)$.

Multiple choice (each question has only one correct answer). You don't need to show your work.

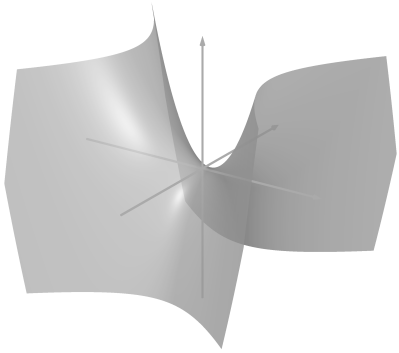
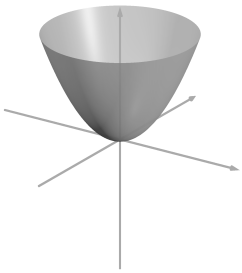
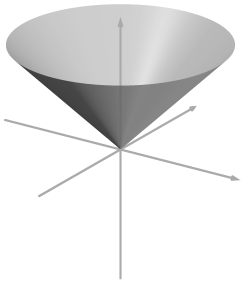
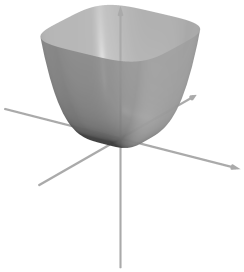
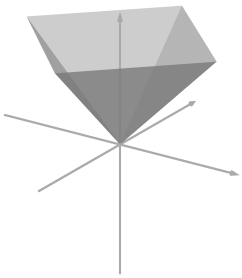
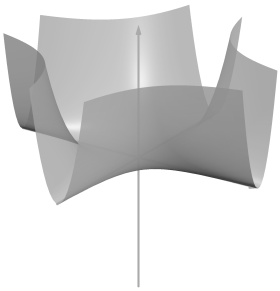
Problem 9 (4 points) Match the following equations to the graphs by writing the corresponding letter below the graph. Two graphs will not be matched.

a) $z = \sqrt{x^2 + y^2}$.

b) $z = x^2 + y^2$.

c) $z = x^2 y^2$.

d) $z = x^4 + y^4$.

In reading order: none, b, a, d, none, c.

Problem 10 (3 points) Let $f(x, y)$ be a differentiable function. Which of the following vectors must be orthogonal to the graph of f at $(a, b, f(a, b))$?

a) $\langle f_x(a,b),\ f_y(a,b),\ 1\rangle$.

b) $\langle f_x(a,b),\ f_y(a,b),\ -1\rangle$.

c) $\langle -f_y(a,b),\ f_x(a,b),\ 1\rangle$.

d) $\langle -f_y(a,b),\ f_x(a,b),\ -1\rangle$.

b). The tangent plane is $z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$, which has normal vector $\langle f_x(a,b), f_y(a,b), -1\rangle$.